The ISR Data Avalanche

India's drone + satellite imagery grows faster than analysts can review it.

40% /yr
imagery growth rate

<20%
frames analyzed by humans

Foreign cloud
AWS / Google / Azure — unacceptable

Closed systems
domestic AI = non-explainable

We built ASTRA.

Sovereign, explainable, edge-deployable AI for aerial target recognition.

Four-Layer Sovereign Stack

Fully offline · pixel-level explainable · human-in-the-loop · robustness-tested

LAYER 4

COMMAND & CONTROL

6-view dashboard · Mission · Map · Alerts · Audit · Benchmark · Adversarial

LAYER 3

FUSION & REASONING

ByteTrack · geo-ref · threat scoring · Grad-CAM explainability · narrator · PDF reports

LAYER 2

INFERENCE

YOLOv11-nano · 2.6M params · 22 ms GPU / 36 ms CPU · 5.5 MB ONNX

LAYER 1

INGEST

file upload · RTSP/HTTP stream · GPS-from-EXIF · satellite tiles

SOVEREIGN

zero foreign API calls · air-gap ready

EXPLAINABLE

Grad-CAM + natural-language narrator

HUMAN-IN-THE-LOOP

no autonomous engagement path

ROBUSTNESS-TESTED

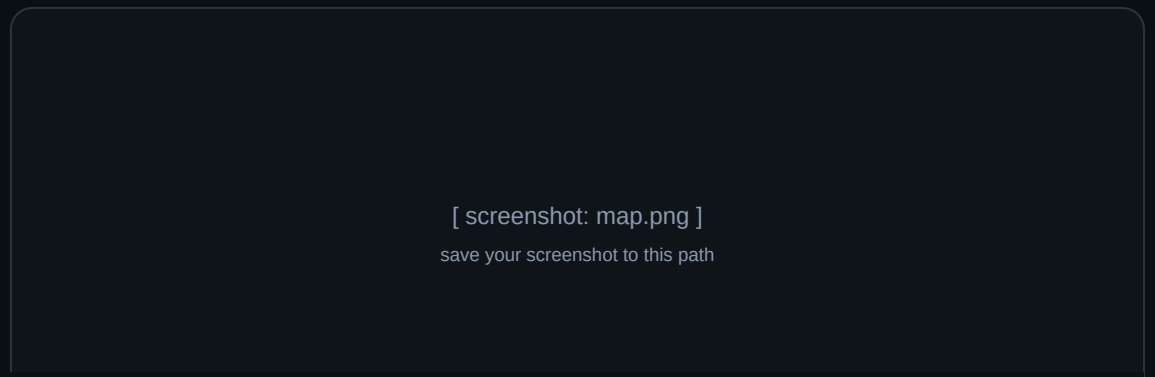
measured adversarial weakness, exposed

Upload · Detect · Explain · Geo-locate · Rank · Audit

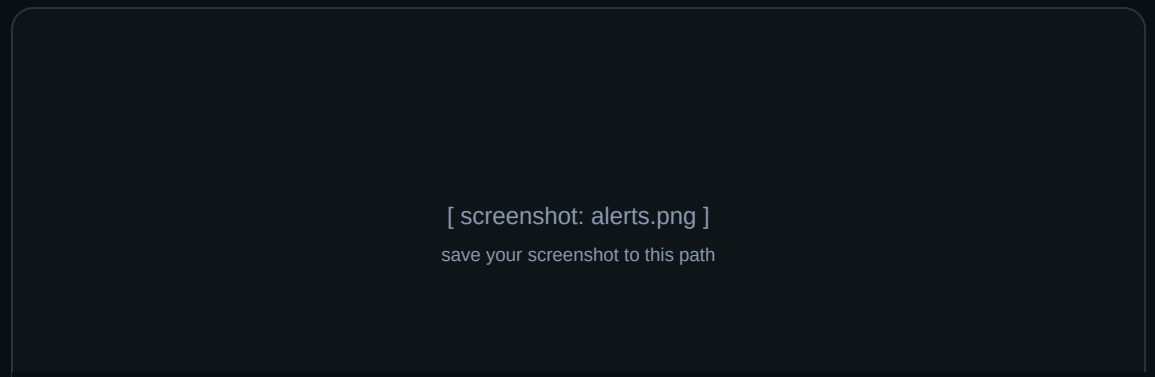
Running at http://localhost:5173 · sovereign, offline, no external calls

Mission · 32 detections in 151 ms

Grad-CAM · pixel-level explainability



Map · real geo-coordinates, AOI box



Alerts · narrator + re-rankable queue

Measured, not marketed.

DOTA-v1.5 held-out val · 1,561 images · 67,029 annotated objects

MAP@0.5 OVERALL
0.515
across 8 classes · 0.915 on aircraft

GPU LATENCY (MEDIAN)
22.3 ms
RTX 4050 Laptop · 44.9 FPS

CPU LATENCY (MEDIAN)
36 ms
Ryzen 7 · ONNX Runtime · 28 FPS

MODEL SIZE
5.5 MB
yolo11n ONNX · Jetson-Nano-ready

[screenshot: benchmark.png]
save your screenshot to this path

ADVERSARIAL ROBUSTNESS — FGSM, measured live

We do not hide weakness. We measure it. Mitigation is roadmap v1.2.

ε = 0.00	100% retained	baseline
ε = 0.02	77% retained	light degradation
ε = 0.05	46% retained	vehicle classes collapse
ε = 0.12	catastrophic	false-positive bloom

Trained in 1h 53m on a ■90,000 consumer laptop. Not cloud.

Roadmap — what ships next

v1.1

2 weeks

MAR20 fine-grained military aircraft
(Su-57, MiG-29, F-16 variants)
rotated-bbox for bridges

v1.2

1 month

Adversarial training
(randomized smoothing)
DJI telemetry → pinhole georef

v2.0

3 months

Multi-camera fusion
movement-intent scoring
federated training across bases

v3.0

6 months

Jetson Orin Nano edge deploy
air-gapped ops
MIL-STD-882 safety case

IGN. EXPLAINABLE. MEASURED.

— ASTRA · BMSIT&M Bengaluru · Apr 30 2026 —